Implementing Sparse State Preparation

Study notes on Rupprecht & Wölk

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We follow Section 2 of [1]. A sparse target state is

$$|\Theta\rangle = \sum_{i=0}^{s-1} c_i \, |C_i\rangle \,,$$

with each $|C_i\rangle$ an n-bit computational basis state and $s \ll 2^n$.

Definition 1.1 (Tableau)**.** *Stack the bitstrings C_i as the rows of an $s \times n$ bit matrix. Let $l = \lceil \log_2 s \rceil$. Split every row at position l into $|C_i\rangle = |C_i'\rangle_s \, |C_i''\rangle_r$, the* subspace register *(l qubits) and the* non-subspace register *($n - l$ qubits).*

The goal is a circuit G and an injective map f with $G\,|C_i\rangle = |f(i)\rangle_s \, |0\rangle_r$ for all i. The state preparation circuit is then the inverse of G applied after a dense preparation on l qubits.

Definition 1.2 (Batch size)**.** $m = \max\{\, 2^p : 2^p \leq n - l \,\}$.

Worked example. The paper's seven states have $s = 7$, $n = 7$, so $l = \lceil \log_2 7 \rceil = 3$ and $m = 4$. The first state is `011|1011`: subspace part `011`, non-subspace part `1011`.

Remark 1.1. Every gate in G (X, CX, Toffoli, multi-controlled X) permutes computational basis states, so G can be simulated on bitstrings alone. This is why the code can be tested classically.

References

[1] Felix Rupprecht and Sabine Wölk. *Sparse quantum state preparation with improved Toffoli cost.* 2026. arXiv: `2601.09388 [quant-ph]`.